

Linear Regression

Exam / Study Cheat Sheet

1. Core Idea

Linear regression models the relationship between a dependent variable y and one or more independent variables x by fitting a straight line (or hyperplane) that minimizes the error between predicted and actual values.

Simple Linear Regression	$y = \beta_0 + \beta_1 x + \varepsilon$
Multiple Linear Regression	$y = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n + \varepsilon$
Matrix Form	$y = X\beta + \varepsilon$

2. Estimating Coefficients — Ordinary Least Squares (OLS)

OLS finds β by minimizing the Residual Sum of Squares (RSS) between actual and predicted y .

$$RSS = \sum (y_i - \hat{y}_i)^2$$

Closed-form (matrix) solution:

$$\beta = (X^T X)^{-1} X^T y$$

For simple linear regression:

$$\beta_1 = \sum (x_i - \bar{x})(y_i - \bar{y}) / \sum (x_i - \bar{x})^2$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

3. Key Assumptions (LINE + others)

- **Linearity** — relationship between X and y is linear
- **Independence** — residuals are independent of each other
- **Normality** — residuals are normally distributed
- **Equal variance (Homoscedasticity)** — residual variance is constant across all X
- No (or low) multicollinearity among predictors
- No significant outliers / high-leverage points

4. Cost Function & Gradient Descent

Mean Squared Error (cost function):

$$J(\beta) = (1/n) \sum (y_i - \hat{y}_i)^2$$

Gradient descent update rule (learning rate α):

$$\beta_j := \beta_j - \alpha \cdot \partial J / \partial \beta_j$$

5. Evaluation Metrics

MAE	$(1/n) \sum y_i - \hat{y}_i $	Avg. absolute error; robust to outliers
MSE	$(1/n) \sum (y_i - \hat{y}_i)^2$	Penalizes larger errors more
RMSE	$\sqrt{\text{MSE}}$	Same units as y ; interpretable
R^2	$1 - (SS_{\text{res}} / SS_{\text{tot}})$	Proportion of variance explained (0–1)
Adj. R^2	$1 - [(1 - R^2)(n - 1) / (n - p - 1)]$	Penalizes adding useless predictors

6. Regularization (Bias-Variance Tradeoff)

Ridge (L2)	$J(\beta) + \lambda \sum \beta_j^2$	Shrinks coefficients, keeps all features
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Lasso (L1)	$J(\beta) + \lambda \sum \beta_j $	Shrinks & can zero-out coefficients (feature selection)
Elastic Net	$J(\beta) + \lambda_1 \sum \beta_j + \lambda_2 \sum \beta_j^2$	Combines L1 + L2

7. Common Pitfalls & Diagnostics

- **Multicollinearity** — check with VIF (Variance Inflation Factor); VIF > 5–10 is concerning
- **Heteroscedasticity** — detect via residual-vs-fitted plot (funnel shape) or Breusch-Pagan test
- **Outliers / leverage** — check Cook's Distance, studentized residuals
- **Overfitting** — high train R^2 but low test R^2 ; use regularization or cross-validation
- **Non-linearity** — fix with polynomial terms, log transforms, or non-linear models

Tip: For exams, be ready to derive β_1 and β_0 for simple linear regression, explain LINE assumptions, and compare R^2 vs Adjusted R^2 vs RMSE.